

AI Money Butler

Product requirements, architecture, development phases, implementation plan, and delivery Gantt chart.

PRODUCT DIRECTION

A calm money decision app.

Kira turns a user's current financial picture into safe daily choices without sacrificing the future they value.

The product promise

Kira is not a generic budgeting dashboard and not a generic chatbot. It is a phone-first money butler that captures real spending, protects important commitments, calculates what is safe, and helps a user choose what to do today.

MVP focus	Primary outcome	Explicit boundary
Cash flow, one savings goal, food plus transport planning, and evidence-backed Butler answers.	A user can make a daily decision and see its effect on their goal before spending.	No money movement, investment recommendations, lending advice, or silent plan changes.

1. Product requirements

The first release must answer one important question exceptionally well: **what can I safely do today without breaking the financial future I already care about?**

Core user journey

Step	User action	Kira outcome
1. Build picture	Enter balance, payday, essential bills, emergency buffer, and a goal.	Creates a validated cash-flow snapshot and data confidence level.
2. Capture reality	Add a manual expense, receipt, voice note, import, or later a read-only account feed.	Creates a reviewable transaction draft and updates the ledger after confirmation.
3. Plan a decision	Ask if a purchase is affordable or request lunch and transport options.	Returns a cost, timing, goal impact, and assumptions - not vague budgeting advice.
4. Replan	Record an overspend or choose a higher-cost option.	Proposes recovery scenarios while keeping rent, debt minimums, and the buffer protected.
5. Approve	Accept, edit, or reject Kira's proposed change.	Writes a versioned plan update only after the user approves.

MVP success criteria

- Safe-to-spend is calculated from confirmed transactions, bills, payday, emergency buffer, and goal contribution.
- Every OCR or voice transaction remains a draft until a user confirms it.
- Butler answers show the facts and constraints used to reach the result.
- A selected food-and-transport option shows total estimated cost, arrival time, and effect on the active goal.
- An overspend produces at least two safe recovery scenarios; the user decides which one to apply.

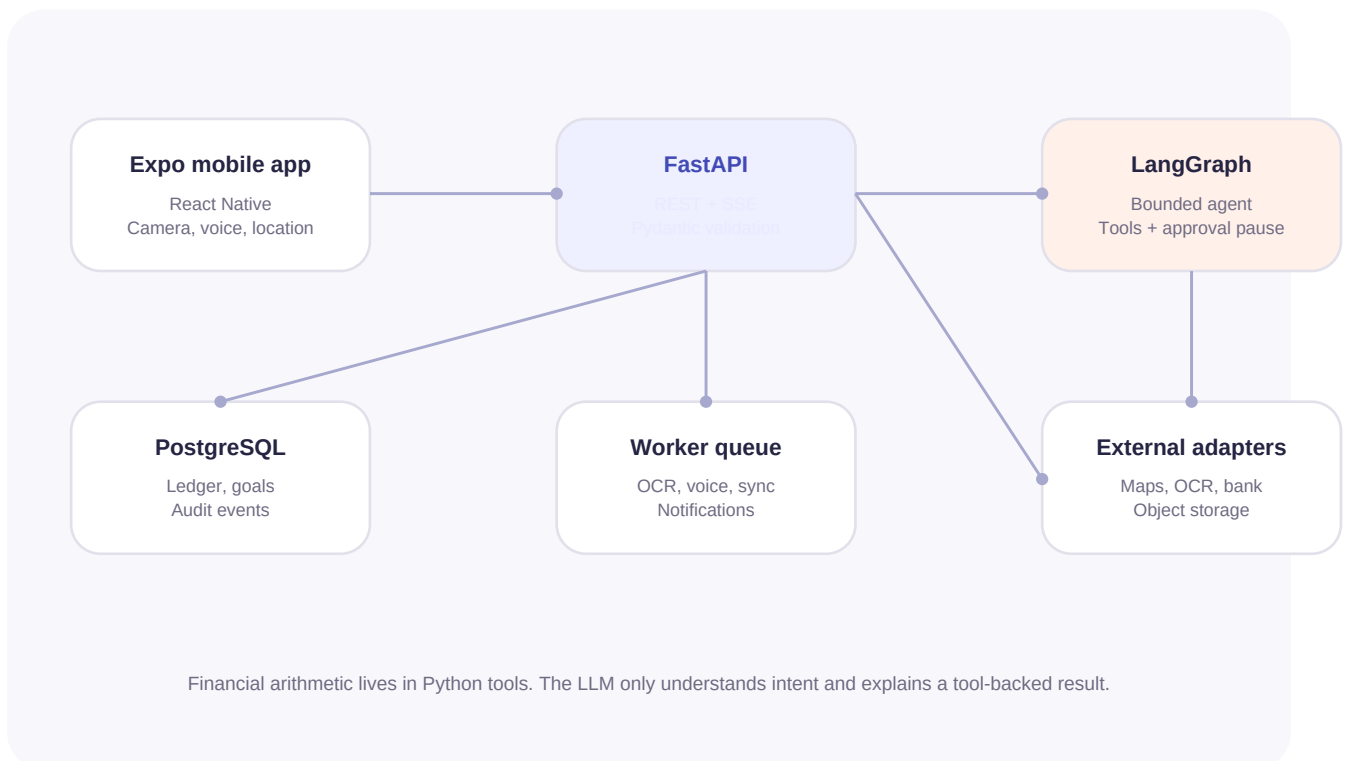
Non-goals for the MVP

Do not build	Reason
Autonomous transfers or payment execution	Requires stronger user trust, banking partnerships, and formal approval controls.
Investment, stock, loan, or credit advice	Creates regulatory and suitability risk outside the budgeting and cash-flow scope.
Exact food pricing promises	Nearby places rarely expose reliable real-time menu prices. Use a clearly labelled estimate.
Generic life-assistant capabilities	Dilutes Kira's differentiation. Expand only into decisions with clear financial consequences.

2. Architecture and technical stack

Kira should be built as a native mobile application, not a browser-first web app. Expo React Native provides the right foundation for camera, microphone, location, secure storage, notification, and Maps handoff while retaining a single TypeScript codebase.

Layer	Recommended choice	Why it belongs here
Mobile	Expo React Native + TypeScript + Expo Router	Phone-first delivery with a custom bottom bar and stable cross-platform navigation.
Mobile state	TanStack Query, React Hook Form, Zod, Expo SecureStore	Separates remote data, validated forms, and protected local session storage.
API	FastAPI + Pydantic v2	Typed REST endpoints, streaming responses, validation, and strong Python fit for solver tools.
Data	PostgreSQL + SQLAlchemy 2 async + Alembic	Durable ledger, transaction provenance, goal history, and auditable plan versions.
Async jobs	Redis + Celery or ARQ worker	Runs OCR, transcription, statement import, notification jobs, and syncs outside user requests.
AI	LangGraph + LangChain tools + provider-agnostic LLM	A bounded assistant that uses typed tools and pauses for sensitive decisions.
External services	Maps adapter, OCR adapter, account-provider adapter, S3-compatible storage	Keeps vendors replaceable and avoids locking product logic to one supplier.



3. Mobile pages and interaction design

Navigation is deliberately simple. The raised middle action is Butler, because asking Kira for a decision is the product's primary interaction - not an afterthought.

Page	Purpose	Required content and action
Onboarding	Create a trustworthy starting picture in under five minutes.	Balances, payday, recurring essentials, buffer, first goal, optional food/transport preferences, explicit permissions.
Today	Show the next best financial decision, not a crowded dashboard.	Safe-to-spend amount, why it is safe, one goal, upcoming commitment, prompt to plan lunch and transport.
Activity	Capture spending with low friction.	Manual entry, receipt scan, voice log, draft confirmation, recent transactions, merchant/category correction.
Plan	Make goals viable and turn them into a daily plan.	Goal progress, payday contribution, protected rules, scenario comparison, day-planning segment.
Butler	Answer financial questions with evidence and a clear recommendation.	Direct answer, assumptions, linked evidence, trade-off, and approval card for any plan-change proposal.
Accounts	Manage read-only balances and import sources.	Connection status, last sync, cash account, transaction import, revoke connection.
Bills	Protect future commitments before daily spending.	Bills, subscriptions, due dates, reserved total, reminder setting, cancellation-review handoff.
Safety	Let users control trust boundaries.	Data confidence, emergency buffer, location permission, approval requirement, data export/delete request.

Bottom navigation

Today - Activity - Butler - Plan - More. Accounts, Bills, and Safety live under More. The design should use one clear action per screen, large whitespace, and no dense super-app dashboard patterns.

4. Functional modules and data model

Module	Responsibilities	Primary records
Identity and profile	Authentication, consent, preferences, timezone, food and transport defaults.	users, sessions, consents, preferences
Ledger	User-owned financial source of truth. Normalise merchant, category, source, amount, date, and status.	accounts, transactions, merchants, categories
Ingestion	Create transaction drafts from a receipt, voice note, statement import, or account adapter.	ingestion_jobs, receipt_files, transcriptions
Commitments	Detect and manage bills, subscriptions, debt minimums, and expected payment dates.	recurring_rules, bills, subscriptions
Goals and finance engine	Calculate safe-to-spend, contribution schedule, feasibility, and recovery scenarios.	goals, protection_rules, goal_plans, plan_scenarios
Day planner	Rank nearby food and route options by cost, timing, preference, and goal impact.	day_plans, place_estimates, route_estimates
Butler and approval	Store threads, tool runs, proposed actions, user decisions, and audit events.	assistant_threads, agent_runs, approvals, audit_events
Notifications	Generate bill reminders, low-confidence notices, and plan-change prompts.	notifications, notification_preferences

4.1 Module and feature matrix

This is the clear build scope. Every feature belongs to one module, has a defined user-facing page, and is implemented through a backend service rather than being left as a generic AI capability.

Module	Features included in MVP	Primary pages
Core financial tracker	Balances; cash account; manual transaction; category and merchant; source tag; transaction history; CSV/statement import.	Onboarding, Today, Activity, Accounts
AI entry	Receipt OCR; voice expense entry; merchant/category suggestion; confidence score; editable review; one-tap confirmation.	Activity, receipt-confirm
Commitment manager	Recurring bill detection; subscription list; due-date reminders; reserved total; debt minimum protection; review-cancellation handoff.	Today, Bills
Finance planner	Safe-to-spend today; safe-to-spend until payday; emergency buffer; goal by date; feasibility; scenarios; overspend replanning.	Today, Plan
Day planner	Current location on demand; destination and deadline; food preferences; walking/public transport/ride estimate; three combined options; Maps deep link.	Plan - Plan my day
Kira Butler	Affordability questions; explain plan changes; upcoming-risk prompt; evidence-backed recommendations; plan-change draft; approve/edit/reject flow.	Butler
Connected data	Read-only account/e-wallet adapter; connection status; last sync; revoke; provider sandbox or statement-import fallback.	Accounts
Trust and control	Data-confidence label; protected buffer; explicit plan-change approval; location consent; audit trail; data export/delete request.	Safety
Notifications	Bill reminder; unresolved receipt/voice draft; low-balance risk; requested approval; goal deviation notice.	Today, Bills, Safety

Data rules

- Store money as integer sen with ISO currency code. Never use floating-point arithmetic for financial calculations.
- Keep a source and confidence value for every transaction and every cost estimate.
- Version every accepted goal plan and preserve the plan version used for each Butler answer.
- Use UTC in storage and the user's timezone only for display, due-date logic, and notification schedules.
- Encrypt provider credentials and keep an audit record for sensitive reads, writes, approvals, and revocations.

5. Backend implementation plan

FastAPI should expose a clean domain API. The mobile application never calculates financial truth locally; it presents data and sends user input. The backend owns ledger integrity, solver results, approval state, and audit history.

Area	Representative endpoints	Important behavior
Dashboard	GET /v1/dashboard/today	Returns safe-to-spend, confidence, upcoming commitment, active goal, and explanation inputs.
Transactions	POST /v1/transactions; PATCH /v1/transactions/{id}	All writes validate money, category, source, and user ownership.
Ingestion	POST /v1/ingestion/receipt; POST /v1/ingestion/voice	Returns draft data; confirmation is separate from extraction.
Accounts	POST /v1/accounts/import; POST /v1/connections; DELETE /v1/connections/{id}	Use provider adapters and explicit read-only consent.
Commitments	GET /v1/bills; POST /v1/subscriptions/{id}/reminder	Reserve expected payments before calculating free cash.
Goals	POST /v1/goals; GET /v1/goals/{id}/plan; POST /v1/goals/{id}/scenarios	Returns deterministic scenarios and plan-change drafts.
Day plan	POST /v1/day-plans; POST /v1/day-plans/{id}/select	Returns price estimate, duration, arrival time, and goal effect.
Butler	POST /v1/butler/runs; GET /v1/butler/runs/{thread_id}/stream	Streams explanation and surfaces approval interrupts.
Approvals	POST /v1/approvals/{id}/respond	Accept, edit, or reject a proposed plan change. Never auto-apply.

Worker jobs

Use the worker queue for OCR, transcription, account refresh, recurring-pattern detection, notification dispatch, and long-running day-plan/map calls. Keep FastAPI requests short and return job status or streamed events for operations that take longer than the user should wait.

6. Butler implementation with LangGraph

AGENT BOUNDARY

The LLM does not own the money.

LangGraph interprets user intent and chooses approved tools.
Python tools calculate. Users approve anything that changes a plan.

Graph stage	Responsibility	Can it write?
1. Intent extraction	Convert user language into typed intent: affordability, explanation, replan, or day plan.	No
2. Policy guard	Reject unsafe scopes; enforce user ID, data scopes, and explicit consent requirements.	No
3. Read tools	Load financial snapshot, bills, transaction evidence, goal, and preferences.	No
4. Deterministic tools	Calculate cash flow, safe-to-spend, goal scenarios, route options, and price estimates.	No
5. Response composer	Explain a tool-backed answer in plain language with assumptions and caveats.	No
6. Approval interrupt	Pause on a proposed plan change and wait for approve, edit, or reject.	Draft only
7. Audit and resume	Apply approved change, log decision, update plan version, and continue the thread.	Yes, after approval

Required tools

Tool	Purpose
get_financial_snapshot	Read confirmed balances, income, bills, buffer, active goal, and data confidence.
calculate_safe_to_spend	Return deterministic disposable amount for today or until payday.
run_goal_scenarios	Model timeline and contribution trade-offs after an event or proposed purchase.
build_day_plan	Combine food and transport candidates, travel time, estimated cost, and goal impact.
create_plan_change_draft	Create a proposed plan change without applying it.
request_user_approval	Create the pause that allows approve, edit, or reject.

Safety rule: tools that move money, cancel a subscription, or write financial data must not be available to the agent without a specific approval policy. The MVP has no money-moving tool at all.

7. Integration and external-service plan

All external systems must sit behind adapters. This lets the product work with sandbox data and statement imports before a bank or e-wallet partnership is available.

Adapter	MVP implementation	Later production path
Receipt OCR	Upload image to worker; return merchant, amount, date, and confidence as a draft.	Provider fallback, receipt image retention policy, and merchant-learning model.
Voice entry	Record short voice note; transcribe to a draft transaction with editable fields.	Malay and English language handling, offline draft capture, and user corrections feedback loop.
Account import	CSV or sample statement import; normalise into a review queue.	Read-only account-provider adapter with explicit consent and encrypted refresh tokens.
Maps	Nearby food candidates plus route duration and a deep link to Maps.	Route-aware place ranking, saved places, better price estimates, and map-provider fallback.
Notifications	Local and push reminders for bills, unresolved drafts, and approval requests.	Personalised timing and user-controlled notification policies.

Food and transport planner

For each request, the backend receives current location, destination, deadline, food preference, transport preference, and current safe-to-spend amount. It fetches candidates, calculates route legs, estimates the full outing cost, and ranks options. The UI shows three choices, never a single imposed answer.

Rank signal	How Kira should use it
Hard constraints	Arrival deadline, dietary preference, selected transport mode, protected commitments, available cash.
Cost estimate	Venue price level, known user spend, local/curated menu estimate; display confidence and estimate label.
Convenience	Walking time, route duration, route changes, and detour from destination.
Goal effect	Show whether the option preserves the active goal, consumes flexible room, or requires a recovery scenario.

8. Development phases and Gantt chart

This is an eight-week MVP plan for one full-stack developer, with a product/design owner validating the flow weekly. It deliberately treats live bank integration as an adapter milestone, not as the dependency that can block the core product.



Phase	Deliverable gate
Week 1	Phone shell, custom navigation, authentication, seed/demo data, onboarding, and Today page.
Week 2	Ledger, manual entry, commitment model, recurring bills, and first safe-to-spend result.
Weeks 3-4	Goal solver, scenario comparison, overspend replanning, and deterministic finance test suite.
Week 5	Receipt and voice draft flows with confirmation; worker queue and notifications.
Week 6	Food plus transport planner, Maps deep link, cost confidence labels, and selection result.
Week 7	Butler graph, typed tools, evidence display, approval interrupt, and audit events.
Week 8	CSV/sandbox account adapter, end-to-end test, demo script, polish, and security review.

9. Testing, quality gates, and demo readiness

Kira wins trust by behaving predictably. The most important testing is not visual polish; it is proving that financial constraints and user-approval boundaries cannot be bypassed.

Area	Must prove before demo
Finance engine	Same inputs always produce the same safe-to-spend amount and scenario outputs. No floating point error. Protected commitments cannot be used.
Ingestion	A bad OCR or voice extraction remains a draft, can be corrected, and never silently alters the ledger.
Butler	Every affordability answer includes its evidence. The agent cannot call an unapproved write tool.
Approval	A proposed replan pauses, can be edited/rejected, and changes the plan only after explicit approval.
Day plan	Displayed price is clearly estimated; route timing and destination are visible; selection updates the proposed budget impact.
Privacy	Location is requested only for a user-triggered day plan. Provider tokens are encrypted and can be revoked.
Mobile UX	All core tasks work on a phone with readable text, clear empty/error states, and no dense dashboard overload.

Competition demo script

- Show the user's balance, payday, rent, phone bill, emergency buffer, and wedding goal on Today.
- Add a voice or receipt transaction and confirm it. Safe-to-spend changes visibly.
- Ask Kira to plan lunch and transport to a destination before a deadline.
- Select the higher-cost option and show the goal impact.
- Open Butler. Kira proposes a recovery scenario that protects rent, debt, and the buffer.
- Approve or reject the proposal to prove that the user remains in control.

Technical references

Expo Router: docs.expo.dev/router/introduction/ | FastAPI: fastapi.tiangolo.com/ | LangGraph human-in-the-loop: docs.langchain.com/oss/python/langchain/human-in-the-loop | Google Maps Platform: developers.google.com/maps/documentation